# Two-Phase PageRank Pipeline

## GPS map-matching, diffusion smoothing, and chain evaluation

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## ABSTRACT

We fit a calibrated, per-link model of urban traffic on a 99,716-link metropolitan road network using 30 days of probe GPS trajectories. The pipeline map-matches the raw fixes with the Newson-Krumm hidden Markov matcher [3], accumulates a sparse popularity matrix $C \in \mathbb{Z}_{\geq 0}^{T \times N}$ with $T = 8{,}640$ five-minute bins, and smooths each row of $C$ by one step of graph diffusion [19] at a strength calibrated to the empirical cross-week noise floor of the corpus. The smoothed prior then drives a two-phase PageRank chain that decomposes a random walk into a peak-seeking up phase of geometric length $L_{\uparrow} \sim \mathrm{Geo}(\beta)$ and a destination-seeking down phase $L_{\downarrow} \sim \mathrm{Geo}(\rho)$, with $(\beta, \rho)$ fit from the empirical trip-length distribution of the corpus rather than chosen by hand. On a random sample of evaluation bins the two-phase chain reconstructs its own input within roughly 73% Top-100 relative error on the busiest links and forecasts the next week's signal at the level of the cross-week noise floor; both numbers improve on the single-phase and naive baselines.



## CCS CONCEPTS

• **Computing methodologies** → **Modeling and simulation**; Spatial and physical reasoning; • **Information systems** → *Geographic information systems.*

## KEYWORDS

PageRank, road traffic, GPS trajectory, map matching, hidden Markov model, graph diffusion, urban mobility, per-link traffic estimation

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## 1 INTRODUCTION

The two-phase PageRank chain operates on a road graph and consumes a per-link distribution $E_b \in \mathbb{R}^N$ summarizing how often each road segment is used during a chosen time window. Producing a clean $E_b$ from raw GPS data requires three steps: matching noisy trajectories to network edges, binning the matched fixes into a sparse popularity matrix $C \in \mathbb{Z}_{\geq 0}^{T \times N}$ whose entry $C_{b,i}$ counts distinct trips on link $i$ in bin $b$, and smoothing each row of $C$ before it enters the chain. The following sections cover these steps in order, then define the chain as a column-stochastic Markov operator and report its evaluation against held-out data.

## 2 RELATED WORK

The question this paper addresses is whether one can fit a calibrated model of vehicle traffic at the resolution of *every single road link* of an entire city (~ 100,000 links), using probe trajectories on a fixed road network, and validate the fit against held-out weeks and a held-out real-world intervention. The closest prior work falls into three groups: agent-based traffic simulators that take demand as an input rather than fit it from data [21, 22]; origin-destination estimators that operate at a coarser zone or node resolution [11, 12]; and PageRank-style random walks applied to urban networks for centrality or coarse congestion prediction [8, 23, 24].

*MATSim and agent-based traffic simulation.* The closest comparable effort in scope is MATSim [21, 22], a long-running agent-based simulator that, given a synthetic population, schedules, and a network, performs *forward* simulation of per-link traffic at every time of day. Our problem is the *inverse*: given observed probe trajectories, infer a per-link popularity field and validate it against held-out weeks. The two efforts complement rather than compete; we use the MATSim network model of Salt Lake City as our graph, but the calibration-from-data step is outside MATSim's scope.

*Origin-destination estimation: a coarser unit of modeling.* A long line of transportation work estimates city-wide origin-destination (OD) matrices from cellular records combined with link counts [11] or from sparse probe trajectories combined with link counts [12]. These methods produce an OD matrix at zone or node resolution: each entry quantifies how many trips begin in zone $a$ and end in zone $b$, but commits to no particular route in between. The per-link traffic field we estimate sits at a strictly finer modeling unit, and OD estimation is a coarser predecessor rather than a direct competitor.

*PageRank applied to road and urban networks.* PageRank has been adapted to urban networks in three lines of work that are distinct from each other and from ours.

*Adapted PageRank as a structural centrality.* [8] apply a non-local random walk to urban street networks to compute a node centrality, working from the topology of the street graph rather than from observed traffic. The output is a structural centrality ranking, not a calibrated traffic field; the random walk has no learned transition or teleportation parameters and is never compared against observed link occupancy.

*PageRank scores as predictors of regional congestion.* [23] partition Beijing into a ~4,000-cell grid, weight inter-cell edges by taxi flow at each time slot, run PageRank per slot, and report a linear relation between a cell's PageRank now and its congestion at the next slot. This is genuinely PageRank-for-traffic, but at a strictly coarser unit



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than ours (4,000 cells vs 99,716 links), with no calibrated random-walk parameters, no held-out validation, and a scalar congestion target per cell rather than a per-link occupancy distribution.

*Weighted PageRank for ranking spaces.* [24] applies weighted PageRank to a space-syntax connectivity graph of axial lines and validates that PageRank scores rank-correlate with observed pedestrian and vehicular movement rates at four observation sites in central London (50-minute counts). The result is a relative ranking of urban spaces, not a calibrated per-link traffic field; there is no probe-data fit of the random walk's transition or teleportation probabilities, no city-scale per-link evaluation, and no held-out time-period validation.

# 3 DATA AND PREPROCESSING

## 3.1 Probe corpus

The raw input is a corpus of sparse GPS trajectory records collected in the Salt Lake City metropolitan area between August 31, 2018 and September 30, 2018, a continuous 30-day observation window. The data are distributed as a set of comma-separated value files (one per collection shard) without a header line. Each row corresponds to a single GPS fix and contains four fields:

| Column | Meaning |
|---|---|
| `did` | anonymized device identifier (string) |
| `ts` | timestamp, formatted `YYYY-MM-DD HH:MM:SS` |
| `lat` | latitude of the GPS fix (decimal degrees) |
| `lon` | longitude of the GPS fix (decimal degrees) |

Rows belonging to the same device are interleaved chronologically across files. Coverage is spatially sparse: at any single short time window only a small fraction of the road network is actually touched by at least one device.

The raw corpus contains 924,379,163 GPS fixes; roughly 80% of them fall outside a generous Salt Lake City bounding box and are dropped at the first preprocessing stage, leaving 179 million fixes inside the metro area. After full preprocessing and the map matching of Section 3.2, the corpus is reduced to approximately 2,477,000 distinct trips, of which about 714,594 survive map matching and contribute to the popularity matrix. The remaining trips are discarded for one of three reasons: fewer than five GPS fixes, fixes that fall outside the road network, or no viable Viterbi path. At this stage the data are purely geographic point clouds: the observations are not yet associated with any road link, and no network structure has been introduced.

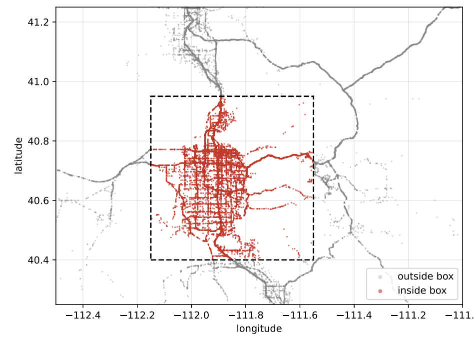


**Figure 1: Raw GPS fixes around the SLC bounding box (random sample of $\sim 10^5$ from 924 M). Red points fall inside the box (40.4–40.95°N, 112.15–111.55°W, dashed); gray fall outside.**

## 3.2 Map-matching

**HAMID:** We project the raw GPS fixes onto the road network with the Leuven implementation [1, 2] of the hidden Markov map-matcher of Newson and Krumm [3]. The matcher models the GPS-to-link perpendicular distance as a zero-mean Gaussian with standard deviation $\sigma$ (in metres); $\sigma$ is the only emission parameter and is corpus-dependent. Newson and Krumm fit $\sigma \approx 4.07$ m on their Seattle corpus; on our sparser SLC corpus we refit by self-consistency. We re-run the matcher across a range of candidate $\sigma$ (with a candidate pruning radius `max_dist` = 400 m, the matcher's default upper bound on emission distance), compute the standard deviation of the per-fix residuals over the surviving fixes, and pick the smallest $\sigma$ at which the declared emission std is no longer smaller than the measured one. **JEFF:** mention their 4.07 m and mention what we did

**Table 1: Self-consistency check for the GPS noise scale $\sigma$ on the SLC corpus.**

| $\sigma_{\text{declared}}$ | $\sigma_{\text{std}}$ (filtered) | gap = $\sigma_{\text{std}} - \sigma$ | verdict |
|---|---|---|---|
| 1.00 | 2.20 | +1.20 | too tight |
| 1.75 | 2.77 | +1.02 | too tight |
| 3.00 | 3.15 | +0.15 | slightly too tight |
| **3.50** | **3.22** | **−0.28** | **first $\sigma > \sigma_{\text{std}}$ (chosen)** |
| 3.75 | 3.25 | −0.50 | too loose |
| 4.00 | 3.26 | −0.74 | too loose |
| 5.00 | 3.31 | −1.69 | too loose |

The sign of $\sigma_{\text{std}} - \sigma_{\text{declared}}$ flips between $\sigma = 3.0$ and $\sigma = 3.5$. We therefore set $\sigma = 3.5$ m throughout this paper.

*Two map versions.* The pipeline operates on two views of the same SLC road network. The *low-resolution* map is the MATSim simulator network ($N \approx 99{,}716$ directed links). The *high-resolution* map is the same OSM extract with long or curved MATSim links subdivided into shorter sub-edges so that arterial geometry is captured by collinear sub-edges. The main pipeline uses high-resolution matching aggregated to low-resolution links; the low-res baseline of Section 9.3 uses low-resolution matching instead.

# 4 POPULARITY MATRIX

The final stage of the pipeline aggregates the matched edge visits into a sparse matrix $C \in \mathbb{Z}_{\geq 0}^{T \times N}$ that is the input to the rest of the project. Each row corresponds to a time bin, each column to a road link, and each entry counts the number of edge visits that fell into that bin and link.

## 4.1 Time bins

Time is partitioned into non-overlapping bins of equal width $\Delta = 300$ seconds (five minutes). For an absolute timestamp $t$ measured in seconds since the epoch $t_0 = 2018\text{-}01\text{-}01$, the bin index is

$$b(t) \;=\; \left\lfloor \frac{t - t_0}{\Delta} \right\rfloor. \tag{1}$$

The binning is global: every matched fix produced anywhere in the 30-day observation window is mapped to the same time grid, which contains $T = 30 \times 24 \times 12 = 8{,}640$ five-minute bins. A five-minute resolution is chosen to balance two competing constraints: too coarse a bin (e.g. one hour) blurs the rush-hour profile of the network; too fine a bin (e.g. one minute) leaves the matrix excessively sparse, with most cells empty even on busy links.

## 4.2 Counting rule

The matched-fix output of Section 3.2 provides, for each trip $r$, a list of pairs $(\mathrm{lid}_j, t_j)$, one per emitting GPS fix. The counting rule is:

(1) **Bin assignment.** Each pair is assigned to the bin

$$b_j \;=\; b(t_j) \;=\; \left\lfloor \frac{t_j - t_0}{\Delta} \right\rfloor. \tag{2}$$

(2) **Per-trip deduplication.** For a given trip $r$, all fixes that fall in the same bin and on the same link are collapsed into a single contribution. Concretely, the set of contributions from trip $r$ is the set of distinct triples

$$\left\{ (b_j,\ \mathrm{lid}_j,\ r) \;:\; j = 1, \ldots, M_r \right\}. \tag{3}$$

If the same trip emits five fixes that all map to the same bin and the same link, that combination is counted exactly once.

(3) **Accumulation.** The popularity-matrix entry $C_{b,i}$ is then the number of distinct trips that contributed at least one fix to bin $b$ on link $i$:

$$C_{b,i} \;=\; \left| \{ r \;:\; \exists\, j \text{ with } b_j = b,\ \mathrm{lid}_j = i \} \right|. \tag{4}$$

The per-trip deduplication ensures that $C_{b,i}$ measures link occupancy by distinct vehicles in the corpus, not the GPS-emission density of those vehicles: a trip that produces several fixes while crossing one link inside a single 5-minute window contributes one unit, not several.

## 4.3 Sparse matrix representation

The matrix $C$ has dimensions $T \times N \approx 8.6 \times 10^8$ cells, where $T = 8{,}640$ is the number of five-minute bins (Section 4) and $N = 99{,}716$ is the number of directed road links. Only a small fraction of these cells are nonzero: at five-minute resolution most road links carry no trip in any specific window. We therefore store $C$ in compressed sparse row format alongside the per-row timestamps and per-column link identifiers, which on the SLC corpus reduces the artifact to roughly 7 MiB.

## 4.4 What $C$ does and does not measure

The matrix $C$ summarizes how many trips, drawn from the 714,594-trip corpus, traversed each road link in each five-minute window of the 30-day observation period. It is a direct empirical proxy for link usage and feeds the prior $E_b$ used in the two-phase PageRank chain.

Two limitations are worth stating up front. First, the corpus is a biased sample of real traffic: it is drawn from a specific population of GPS-emitting devices and is not a uniform sample of all vehicles on the road. Links that systematically attract devices in the sample (e.g. commercial freeway corridors) are over-represented relative to links that attract a different population (e.g. residential cul-de-sacs). The matrix should therefore be read as a *prior* on link usage, not as ground-truth volume. Second, even within the corpus, sparsity is structural: at any single five-minute slot, the median number of GPS fixes per link is zero, and only a few hundred to a few thousand links carry any matched visit at all. To quantify this at the bin-of-week level, we aggregate the first four full weeks of the corpus (8,064 bins, i.e. $4 \times 7 \times 288$ five-minute slots) and ask, for each *(link, bin-of-week)* cell, in how many of the four weeks it received at least one matched fix. The result is in Table 2.

**Table 2: How often each (link, bin-of-week) cell was zero across the four weeks. The cell is structurally empty if all four weeks were zero. "Active" cells are the union of $k = 0, 1, 2, 3$ rows ($5.95 \times 10^6$ in total, about 3% of all cells). The total cell count $201{,}027{,}456 = N \times 2{,}016$ counts *(link, bin-of-week)* pairs for a single week ($2{,}016 = 7 \times 288$ five-minute slots in a week); each cell carries four observations, one per calendar week.**

| weeks with zero count | cell count | fraction |
|---|---|---|
| $k = 0$ (active in every week) | 304,983 | 0.15% |
| $k = 1$ | 499,625 | 0.25% |
| $k = 2$ | 1,153,583 | 0.57% |
| $k = 3$ | 3,994,889 | 1.99% |
| $k = 4$ (never observed in 4 weeks) | 195,074,376 | 97.04% |
| total | 201,027,456 | 100% |

97% of all (link, bin-of-week) cells are zero in every one of the four weeks; only 0.15% are active in all four. Most of the matrix is therefore not just sparse but *structurally empty*, and most "active" cells are active in only one of the four weeks, i.e. represent a single noisy sample rather than a stable rate. The downstream graph-diffusion smoothing described elsewhere in the project addresses this by spreading mass to network neighbors of observed links.

# 5 BUILDING THE TELEPORTATION PRIOR

## 5.1 From counts to a prior by row-normalization

The matrix $C$ is converted into a per-bin teleportation prior by row-normalization. For every bin $b$ with $\sum_i C_{b,i} > 0$, define

$$E_b(i) \;=\; \frac{C_{b,i}}{\sum_j C_{b,j}}. \tag{5}$$

This makes $E_b$ a probability distribution over the $N$ links of the network: the share of trip-presence mass during bin $b$ that landed on link $i$. Bins with no observed trips are assigned the uniform distribution $E_b(i) = \frac{1}{N}$, which acts as an uninformative fallback.

The construction of $E_b$ is the entire purpose of the pipeline described above: every step from raw GPS in Section 3 to the sparse matrix in Section 4 contributes a piece of the journey from a device-level point cloud to a link-level distribution $E_b$ that the smoothing of Section 5.2 regularizes and the model of Section 8 consumes.

## 5.2 Diffusion smoothing of the popularity prior

The teleportation prior $E_b$ summarizes what was actually observed on the road network during bin $b$, but it has two structural shortcomings for predictive use. First, $E_b$ is sparse: a typical 5-minute window puts non-zero mass on only a few thousand of the $N$ links, leaving the rest of the network at exact zero. Second, the empirical signal is noisy at fine resolution: the same heavy corridor can be touched by zero observed trips in one bin and several trips in the very next, simply because no GPS device happened to emit a fix on that link inside that 5-minute slot.

We measure the scale of this noise from the four-week structure of the corpus. For each (link, bin-of-week) cell that the corpus saw at least once, we compute the standard deviation of its normalized count $E_b(i) = C_{b,i}/\sum_j C_{b,j}$ across the four weeks; this captures how much the same link in the same time-of-week jitters from week to week. We call this per-cell quantity $\tau_{b,i}$.

**JEFF:** State the units explicitly: $E_b(i)$ is the probability that a GPS fix falling in bin $b$ lands on link $i$. The figure axis and the text should both say that.

**HAMID:** Concretely, $E_b(i) = C_{b,i}/\sum_j C_{b,j}$ is the fraction of all GPS fixes recorded in bin $b$ that fell on link $i$, i.e. the probability that a single fix drawn uniformly from that 5-minute window lands on link $i$. The cross-week deviation $\tau_{b,i}$ is the spread of that probability across the four weeks at the matched bin-of-week; both quantities are dimensionless probability mass.

The median over the $5.95 \times 10^6$ active cells of Table 2 is $\tau_{\text{global}} = 2.65 \times 10^{-4}$, a single corpus-wide scalar used as a global yardstick. Differences in $E_b$ smaller than $\tau$ are not distinguishable from week-to-week sampling noise; the smoothing step below is calibrated to this scale.

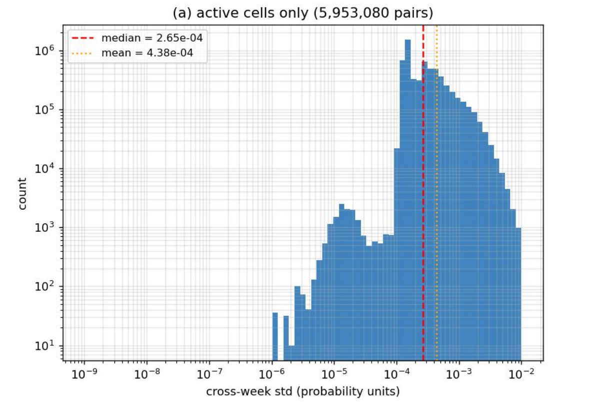


**Figure 2: Distribution of the cross-week standard deviation $\tau_{b,i} = \text{std}_w\, E_b^{(w)}(i)$ across the four weeks, per (link, bin-of-week), restricted to the $5.95 \times 10^6$ active cells. HAMID: Units on the horizontal axis are dimensionless probability mass: a GPS-fix-on-link probability $E_b(i) = C_{b,i}/\sum_j C_{b,j}$.**

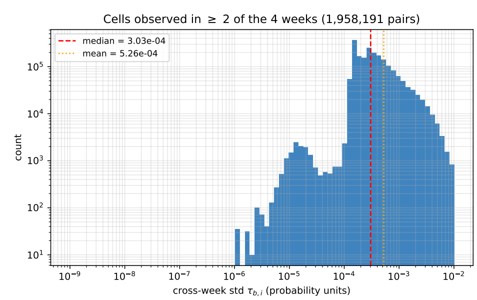


**Figure 3: [temporary] Same distribution as Fig. 2 but restricted to cells observed in *at least 2* of the 4 weeks ($1.96 \times 10^6$ pairs). Removing the one-week-only cells (whose std collapses to a near-zero spike) exposes the actual $\tau_{b,i}$ distribution.**

A graph-aware smoothing step addresses both issues at once. It lifts every link to a positive value (so the prior never has hard zeros) and borrows mass from neighboring links (so a heavy link does not disappear because of an empty bin). The smoothing is applied row by row to the count matrix $C \in \mathbb{Z}_{\geq 0}^{T \times N}$ in three steps.

*Laplace pseudo-count.* For every bin $b$ and every link $i$, replace the raw count by

$$c_b(i) \;=\; C_{b,i} + \alpha, \qquad \alpha = 10^{-2}. \tag{6}$$

After this step every entry of $c_b$ is positive: the row is dense before any further processing.

*Single-step graph diffusion.* Let $P \in \mathbb{R}^{N \times N}$ be the row-stochastic adjacency matrix of the directed road graph: $P_{i,j} = \frac{1}{d_i^{\text{out}}}$ when there is a road link from $i$ to $j$, and zero otherwise; here $d_i^{\text{out}}$ is the out-degree of link $i$. Apply one lazy random-walk step on the graph,

$$\tilde{c}_b \;=\; (1-\gamma)\,c_b \;+\; \gamma\,P\,c_b, \qquad \gamma = 0.20, \tag{7}$$

where $(P\,c_b)_i = \sum_j P_{i,j}\,c_b(j)$ is the average of $c_b(j)$ over the out-neighbors $j$ of link $i$. The mass on link $i$ is therefore averaged with the mass on the links it leads to. Finally, normalize each row to a probability distribution:

$$\tilde{E}_b(i) \;=\; \frac{\tilde{c}_b(i)}{\sum_j \tilde{c}_b(j)}. \tag{8}$$

The result $\tilde{E}_b \in \mathbb{R}^N$ is the *diffused prior*: dense, positive on every road link, and preserving the order of the heaviest links (the top-100 link sets of $E_b$ and $\tilde{E}_b$ overlap on roughly 90% of links across the corpus). $\tilde{E}_b$ is used as an alternative to the raw $E_b$ throughout the evaluation of Section 9.1.

*Choosing $\gamma$ from the noise floor.* The diffusion strength $\gamma$ is not free. We pick it so that the smoothing displaces any single cell by no more than half of that cell's own week-to-week sampling noise: any larger move and the smoother is overwriting structure with neighbor mass rather than filling in noise. Concretely, for each (link $i$, bin-of-week $b$) cell with at least one observed positive count across the four full corpus weeks $w = 1, \ldots, W$, define the per-week displacement

$$D^{(w)}_{b,i}(\gamma) \;=\; \left|\tilde{E}^{(w)}_{b,i}(\gamma) - E^{(w)}_{b,i}\right|, \tag{9}$$

the cell-level displacement averaged across the four weeks

$$D_{b,i}(\gamma) \;=\; \frac{1}{W}\sum_{w=1}^{W} D^{(w)}_{b,i}(\gamma), \tag{10}$$

and compare it to the cross-week noise floor $\tau_{b,i}$ of the same cell. Both $D_{b,i}$ and $\tau_{b,i}$ are in probability units, so the ratio $D_{b,i}/\tau_{b,i}$ is *dimensionless*: "displacement measured in units of one noise-floor unit." The criterion is

$$D_{b,i}(\gamma) \;\le\; \tfrac{1}{2}\,\tau_{b,i}. \tag{11}$$

We compare each cell either against its own cross-week noise scale $\tau_{b,i}$ (*per-cell criterion*, one number per active cell) or against the corpus-wide median noise scale $\tau_{\text{global}} = 2.65{\times}10^{-4}$ (*global criterion*, a single scalar shared by every cell):

$$\text{per-cell:}\quad D_{b,i}(\gamma) \;\le\; \tfrac{1}{2}\,\tau_{b,i}, \tag{12}$$

$$\text{global:}\quad D_{b,i}(\gamma) \;\le\; \tfrac{1}{2}\,\tau_{\text{global}} \;=\; 1.325\times10^{-4}. \tag{13}$$

On the active cells of the SLC corpus ($n_{\text{active}}$ = 5,954,929, per-cell $\tau$ median = $2.3\times10^{-4}$), Table 3 reports three quantiles of each ratio: the median, the 95th percentile, and the fraction of cells whose ratio exceeds the half-noise threshold.

**Table 3: Quantiles of the dimensionless displacement-to-noise ratio $D_{b,i}(\gamma)/\tau$ across the 5,954,929 active SLC cells; left block uses the per-cell denominator $\tau_{b,i}$, right block uses the global denominator $\tau_{\text{global}} = 2.65\times10^{-4}$. Read each row as "at this $\gamma$, half / 95% / fraction-above-0.5 of active cells are displaced by at most so-many noise units." The criterion is the largest $\gamma$ whose 95th-percentile ratio stays $\le$ 0.5. JEFF: remove some of the columns**

| | per-cell criterion | | | global criterion | | |
|---|---|---|---|---|---|---|
| $\gamma$ | med. | 95th pct. | > 0.5 | med. | 95th pct. | > 0.5 |
| 0.00 | 0.000 | 0.000 | 0.00% | 0.000 | 0.000 | 0.00% |
| 0.05 | 0.038 | 0.100 | 0.42% | 0.034 | 0.215 | 0.90% |
| 0.10 | 0.069 | 0.186 | 1.00% | 0.062 | 0.393 | 3.31% |
| 0.12 | 0.081 | 0.219 | 1.27% | 0.073 | 0.463 | 4.40% |
| 0.13 | 0.087 | 0.236 | 1.41% | 0.079 | 0.498 | 4.96% |
| 0.14 | 0.093 | 0.253 | 1.56% | 0.084 | 0.533 | 5.53% |
| 0.15 | 0.099 | 0.270 | 1.73% | 0.090 | 0.568 | 6.10% |
| **0.20** | **0.130** | **0.354** | **2.62%** | **0.117** | **0.741** | **8.95%** |
| 0.25 | 0.160 | 0.436 | 3.63% | 0.144 | 0.912 | 11.87% |
| 0.30 | 0.190 | 0.518 | 5.39% | 0.171 | 1.081 | 14.81% |
| 0.40 | 0.248 | 0.680 | 9.42% | 0.224 | 1.413 | 20.64% |
| 0.50 | 0.306 | 0.838 | 17.29% | 0.276 | 1.737 | 26.59% |
| 0.70 | 0.418 | 1.145 | 32.10% | 0.376 | 2.364 | 37.83% |

**Table 4: [temporary] Displacement of zero cells ($n_{\text{zero}} = 1.95\times10^8$) in units of $\tau_{\text{global}} = 2.65\times10^{-4}$.**

| $\gamma$ | median | 95th pct. | > 0.5 |
|---|---|---|---|
| 0.00 | 0.0004 | 0.0004 | 0.00% |
| 0.05 | 0.0004 | 0.0004 | 0.01% |
| 0.10 | 0.0004 | 0.0004 | 0.03% |
| 0.12 | 0.0004 | 0.0004 | 0.04% |
| 0.13 | 0.0004 | 0.0004 | 0.04% |
| 0.14 | 0.0004 | 0.0004 | 0.05% |
| 0.15 | 0.0004 | 0.0004 | 0.05% |
| **0.20** | **0.0004** | **0.0004** | **0.08%** |
| 0.25 | 0.0004 | 0.0004 | 0.10% |
| 0.30 | 0.0004 | 0.0004 | 0.13% |
| 0.40 | 0.0004 | 0.0004 | 0.20% |
| 0.50 | 0.0004 | 0.0004 | 0.28% |
| 0.70 | 0.0003 | 0.0004 | 0.46% |

Reading the zero-cell table. The median and 95th-percentile ratios are essentially constant at $\approx 4\times10^{-4}$ across all $\gamma$: at least 95% of zero cells never see diffusion mass, they only carry the Laplace floor $\alpha/N$. The fraction of zero cells exceeding the half-noise tolerance is 0.08% at $\gamma = 0.20$ ($\sim 1.5\times10^5$ cells out of $1.95\times10^8$); these are the cells immediately adjacent to active links and they are the ones that get visibly filled in. The takeaway is that the diffusion is bounded even on the cells where the per-cell criterion of Table 3 is undefined: it lifts a small adjacency-defined tail of zero cells into the noise band and leaves the rest at the Laplace floor.

*Choice of $\gamma$.* Under the per-cell criterion the 95th-percentile ratio first crosses 0.5 between $\gamma = 0.25$ (0.436) and $\gamma = 0.30$ (0.518), so

$\gamma^{\mathrm{pc}}_{\max} \approx 0.25$. The stricter global criterion crosses between $\gamma = 0.13$ (0.498) and $\gamma = 0.14$ (0.533), giving $\gamma^{\mathrm{global}}_{\max} \approx 0.13$. We use $\gamma = 0.20$ throughout the paper: it sits comfortably inside the per-cell ceiling, the median active cell is displaced by only 0.117 $\tau_{\mathrm{global}}$ even under the global criterion, and the 95th-percentile ratio crosses the half-noise bound only mildly (0.741).

*Violator tail at $\gamma = 0.20$.* To see how far past the tolerance the violators sit, we sweep the threshold $\eta$ in $D_{b,i}(0.20) > \eta\, \tau_{\mathrm{global}}$ across the active cells:

| $\eta$ | 0.2 | 0.5 | 0.8 | 1.0 | 1.5 | 2.0 |
|---|---|---|---|---|---|---|
| fraction $> \eta\, \tau_{\mathrm{global}}$ | 28.5% | 8.9% | 4.4% | 3.0% | 1.3% | 0.6% |

The violator tail thins quickly: of the 8.9% that exceed half the noise floor, only 3% exceed $\tau_{\mathrm{global}}$ and fewer than 0.7% exceed $2\,\tau_{\mathrm{global}}$, so the smoothing rarely changes a cell by more than its own week-to-week jitter. The per-cell and global criteria agree on the median and differ only on the upper tail (the per-cell version self-normalizes against busy cells, the global version measures every cell against the corpus median); we report both and use $\gamma = 0.20$ as the working compromise.

# 6 EVALUATION FRAMEWORK

We now fix the metrics and protocol used by every subsequent evaluation, including the single-phase baseline of Section 7 and the two-phase chain of Section 8.

## 6.1 Error measures

To evaluate a chain we compare two probability vectors over the $N$ road links: a *truth* vector $t$ and a *prediction* vector $p$, both summing to one (each entry is the share of trip-presence mass on one link in a given bin). Two scalar summaries are used.

*Mean Relative Error (MRE)..* For a numerical regularizer $\varepsilon \geq 0$, define the per-link relative error

$$r_i \;=\; \frac{|t_i - p_i|}{t_i + \varepsilon}, \tag{14}$$

which is dimensionless: numerator and denominator are both in probability units, so $r_i$ measures error as a fraction of the truth itself. We aggregate $r_i$ over two index sets:

$$\mathsf{MRE}_{\text{top-100}} = \frac{1}{100} \sum_{i \in \mathcal{T}_{100}(t)} r_i, \tag{15}$$

$$\mathsf{MRE}_{\text{overall}} = \frac{1}{N} \sum_{i=1}^{N} r_i, \tag{16}$$

where $\mathcal{T}_{100}(t)$ is the set of 100 link indices with the largest values of $t$. The Top-100 MRE focuses on the heavy-traffic head of the distribution, the overall MRE averages across every link in the network. The regularizer $\varepsilon$ is needed only when $t$ has links with $t_i = 0$: it prevents division by zero and acts as a numerical floor that controls how much empty-truth links inflate the average. We report $\varepsilon = 10^{-6}$ throughout, and additionally $\varepsilon = 0$ in configurations where the diffused prior $\tilde{E}_b$ guarantees $t_i > 0$ on every link.

**HAMID:** To anchor the scale, Table 5 summarizes the number of matched GPS fixes in each 5-minute bin across the 8,640 bins of the corpus. The busiest bin contains $M_{\max} = 20{,}881$ fixes, so the smallest possible nonzero per-link probability anywhere in the corpus is $1/M_{\max} \approx 4.8 \times 10^{-5}$. With the uniform per-link probability $1/N \approx 1.00 \times 10^{-5}$ and the empirical noise floor $\tau_{\mathrm{global}} = 2.65 \times 10^{-4}$, our $\varepsilon = 10^{-6}$ sits below all three: it stabilizes the denominator for links with $t_i = 0$ without otherwise perturbing the links we care about. **JEFF:** why $\varepsilon = 10^{-6}$

**Table 5: Matched GPS fixes per 5-minute bin across the 8,640 bins of the corpus.**

| statistic | value |
|---|---|
| total matched fixes | 58,410,362 |
| number of bins | 8,640 |
| zero-fix bins | 60 |
| min | 0 |
| 10th percentile | 1,372 |
| median | 4,791 |
| mean | 6,729 |
| 90th percentile | 14,746 |
| 99th percentile | 17,303 |
| max | 20,881 |

*Three comparisons.* For every target bin $t$ in the evaluation sample we compute the chain's prediction $\hat{v}(t)$ by running the iteration (23) until convergence with $E_b(t)$ (or $\tilde{E}_b(t)$) as the teleportation prior. We then form three (truth, prediction) pairs:

(1) **dataset → dataset** (week to next week): truth = $E_b(t + 7\text{ days})$, prediction = $E_b(t)$. Reports the natural week-to-week variability of the empirical signal at the same weekday and hour-of-day; serves as the *noise floor* the model needs to beat.
(2) **dataset → model** (forecast next week): truth = $E_b(t + 7\text{ days})$, prediction = $\hat{v}(t)$. Reports how well the chain, fed this week's prior, predicts next week's empirical signal.
(3) **dataset → model** (reconstruct this week): truth = $E_b(t)$, prediction = $\hat{v}(t)$. Reports how well the chain reconstructs its own input bin.

With the diffused prior, the same three comparisons replace $E_b$ by $\tilde{E}_b$ on *both sides*: the truth vector, the prediction vector when the prediction is itself a prior (comparison 1), and the chain input that produces $\hat{v}(t)$ (comparisons 2 and 3) are all diffused before the MRE is taken. This keeps every comparison apples-to-apples within a single choice of prior.

*Sampling protocol.* For each of the seven days of the week (Mon, . . ., Sun) we draw 120 source bins uniformly at random, for a total of $7 \times 120 = 840$ target bins. A bin is eligible only if both the bin itself and its +7-day sibling exist in the matrix. In our 30-day corpus this restricts source bins to days 1–23 (Sept 1 to Sept 23) paired with siblings in days 8–30 (Sept 8 to Sept 30); the source bins therefore span weeks 1–3 and the first days of week 4, not just the first week. The seed is fixed (7), so the sample is identical across configurations.

# 7 BASELINE: SINGLE-PHASE PAGERANK

The most natural baseline on the popularity prior $E_b$ is the single-phase PageRank chain with $E_b$ as the teleportation distribution.

Given the row-stochastic adjacency $P$ of the directed road graph (each row of $P$ being uniform over the outgoing road neighbors of one link) and a teleport probability $p \in (0, 1)$, the iteration is

$$v_{n+1} \;=\; p\,E_b \;+\; (1-p)\,P^\top v_n, \tag{17}$$

applied until $\|v_{n+1}-v_n\|_1 < 10^{-7}$, with a safety cap of 300 iterations. The cap is conservative: the power iteration converges geometrically at rate $1-p$, so the theoretical worst-case number of iterations to reach L1 error $10^{-7}$ is

$$n^\star \;=\; \frac{\ln 10^{-7}}{\ln(1-p)} \;=\; \frac{\ln 10^{-7}}{\ln 0.9492} \;\approx\; 310, \tag{18}$$

which sits just above 300. We checked empirically on 1,500+ random target bins and the chain converged before 300 iterations in every single case; the observed maximum was 254 iterations and the median was 209. The cap is therefore never hit in practice and the recorded results are at full $10^{-7}$ tolerance throughout. At every step the walker teleports back to the prior with probability $p$ and otherwise takes a uniform random step along an outgoing road edge. The stationary distribution $v^\star$ of (17) is the single-phase prediction for bin $b$.

*Calibrating p from the trip corpus.* The chain in (17) is a geometric absorption process: the number of graph steps before a teleport is $\mathrm{Geom}(p)$ on $\{0, 1, 2, \ldots\}$ with mean $\frac{1-p}{p}$. We match $p$ to the empirical mean trip length $\mathbb{E}[K_{\text{total}}]$ by method-of-moments, exactly as the two-phase chain matches $\beta$ and $\rho$ to the per-phase lengths:

$$p \;=\; \frac{1}{1+\mathbb{E}[K_{\text{total}}]}. \tag{19}$$

On the noise=3.5 matched corpus (714,594 trips, all route lengths), the empirical mean is $\mathbb{E}[K_{\text{total}}] = 18.68$, which gives $p = 0.0508$. The chain has only this one free parameter; the "damping factor" $1-p = 0.9492$ that some authors report is just a relabeling. Figure 4 shows the empirical $K_{\text{total}}$ histogram with the $\mathrm{Geom}(p = 0.0508)$ reference overlaid.

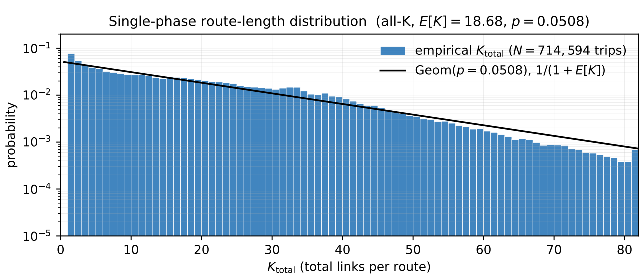


**Figure 4: Empirical histogram of trip length $K_{\text{total}}$ on the 714,594-trip corpus, with the matched $\mathrm{Geom}(p = 0.0508)$ overlay.**

*Why this is a fair baseline.* Three reasons. First, single-phase PageRank is the standard graph-walk model on top of which the two-phase chain is built, so the two share a graph, a prior $E_b$, and a power-iteration solver; the only structural difference is the absence (single-phase) or presence (two-phase) of an up→down split. Second, the teleport parameter $p$ is calibrated from the same trip corpus by the same method-of-moments estimator that yields $\beta$ and $\rho$, so the comparison is not penalized by an artificially detuned baseline. Third, the baseline runs on the same 840 target bins, the same metric, and the same three comparisons as the two-phase chain, per the evaluation contract.

## 7.1 Why single-phase fails on our data

Evaluating (17) on the same 840 target bins as Table 8 (120 per weekday, seed 7, all three eps configurations) gives the numbers in Table 6.

**Table 6: Single-phase PageRank baseline (chain (17) with $p = 0.0508$) on 840 target bins (120 per weekday, seed 7). Conventions match Table 8.**

| | No diffusion $\varepsilon=10^{-6}$ | | Diffused $\varepsilon=10^{-6}$ | | Diffused $\varepsilon=0$ | |
|---|---|---|---|---|---|---|
| Comparison | T-100 | Ov. | T-100 | Ov. | T-100 | Ov. |
| 1. dataset→dataset (next week) | 0.626 | 5.82 | 0.586 | 0.68 | 0.586 | 0.90 |
| 2. dataset→model (forecast) | 0.841 | 8.70 | 0.813 | 1.07 | 0.814 | 1.46 |
| 3. dataset→model (reconstruct) | 0.792 | 8.07 | 0.761 | 0.90 | 0.762 | 1.22 |

Three readings of Table 6, paired against the two-phase numbers in Table 8.

*The forecast loses to the noise floor.* Forecast Top-100 is 0.813, well above the noise-floor Top-100 of 0.586. The single-phase chain therefore fails the contract's pass criterion that forecast must not be worse than last-week-at-the-same-slot. The two-phase chain in Table 8 is at forecast Top-100 = 0.788, only marginally better than the single-phase baseline at this evaluation.

*Reconstruction is comparable.* The reconstruction comparison probes whether the chain returns the input distribution after running to steady state. With the single-phase chain at $p = 0.0508$ the reconstruction Top-100 is 0.761. The two-phase chain on the same prior reaches reconstruction Top-100 = 0.729, only a small improvement over the single-phase baseline.

*Why the single-phase mixes too much.* At $p = 0.0508$ the chain takes on average $\frac{1}{p} = 19.7$ graph steps before teleporting back to $E_b$, which is the empirical mean trip length, so the chain is calibrated to the right time scale. But 19.7 uniform random steps on the SLC road graph are enough to delocalize the walker's distribution across the network: the steady state of a random walk on a connected graph is the degree-proportional distribution, and even at finite damping the chain has visibly relaxed away from $E_b$ on the heavy links. The two-phase split absorbs that mixing into the down phase ($\rho = 0.101$ leak) while keeping the up phase close to $E_b$ ($\beta = 0.102$ leak, applied on $P^\uparrow$ rather than the uniform $P$), which is why the two-phase reconstruction in Table 8 preserves the heavy-traffic head where the single-phase reconstruction does not. The next section formalizes this two-phase chain.

# 8 TWO-PHASE PAGERANK CHAIN

The two-phase chain of this section turns a popularity prior (the raw $E_b$ from Section 4 or the diffused $\tilde{E}_b$ from Section 5.2) into a per-link prediction by running a discrete-time Markov chain on a road-network state space. Its design departs from a single-phase chain (one state per link) in one important respect: each road link $i \in \{1, \dots, N\}$ is represented by *two* states, an *up-phase* state $i^{\uparrow}$ and a *down-phase* state $i^{\downarrow}$, so the chain has $2N$ states in total. The doubled state space lets the chain track two distinct regimes that a vehicle on the network can occupy, and lets each regime contribute its own characteristic mass to the link-level prediction.

## 8.1 What the two phases represent

The two phases of the chain correspond to two qualitatively different regimes of a vehicle's behavior on the road network:

- The *up phase* represents a vehicle that has just appeared in the network and is choosing where to go: it has not yet committed to traveling along a specific corridor. Steps within the up phase let mass redistribute along the graph before any actual travel begins.
- The *down phase* represents a vehicle that is *traveling* along the network, advancing from one link to the next according to road geometry. Steps within the down phase trace out the kind of multi-link path that a real trip would follow.

A single-phase Markov chain on the $N$ physical links cannot distinguish these two regimes: it would force a single transition rule to govern both "where does mass come from" and "where does mass travel to", and any stationary distribution it produced would fold the two contributions into one number per link. The two-phase chain keeps them separate, so that the steady-state mass on each physical link decomposes into a contribution from up-phase deliberation and a contribution from down-phase travel, both controlled by their own transition rule. The next subsections specify those transition rules and the parameters that govern them.

## 8.2 Transition rule

The state space is $\{1, \dots, N\} \times \{\uparrow, \downarrow\}$: every road link $i$ is duplicated into an up-state $i^{\uparrow}$ and a down-state $i^{\downarrow}$. A mass vector on this space is a probability over $2N$ entries (it sums to one), and the chain's stationary distribution decomposes naturally into an up-phase share $v^{\uparrow}(i)$ and a down-phase share $v^{\downarrow}(i)$ per link.

The chain is fully specified by the rule that takes a current state to its next state. From an up-phase state $i^{\uparrow}$ the rule is

- with probability $1 - \beta$, the chain stays in the up phase and moves to one of $i$'s graph neighbors $j^{\uparrow}$, with the choice governed by an up-phase transition matrix $P^{\uparrow} \in \mathbb{R}^{N \times N}$;
- with probability $\beta \in (0, 1)$, the chain commits to traveling along link $i$ and moves to the down-phase state $i^{\downarrow}$ on the same link.

From a down-phase state $i^{\downarrow}$ the rule is

- with probability $1-\rho$, the chain stays in the down phase and moves to a graph neighbor $j^{\downarrow}$, governed by a down-phase transition matrix $P^{\downarrow} \in \mathbb{R}^{N \times N}$;
- with probability $\rho \in (0, 1)$, the trip ends and the chain teleports back to a fresh up-phase state drawn from the prior $E_b$: it jumps to $j^{\uparrow}$ with probability $E_b(j)$.

We adopt the *column-stochastic* convention throughout: column $i$ of any of these matrices stores the probability distribution over next states, given that the chain is currently in state $i$. In particular, $P^{\uparrow}$ and $P^{\downarrow}$ are column-stochastic matrices supported on the directed road graph: the entry $P^{\uparrow}_{j,i}$ is the probability of moving to link $j$ from link $i$ in the up phase, and is zero whenever there is no road link from $i$ to $j$ (and similarly for $P^{\downarrow}$). Below, $v$ denotes the chain's mass vector at iteration step $n$; like $E_b$, it is a probability over the $2N$ states (so $\sum_i v^{\uparrow}(i) + \sum_i v^{\downarrow}(i) = 1$). The chain's prediction for link $i$ is the sum $v^{\uparrow}(i) + v^{\downarrow}(i)$, itself a probability over the $N$ links.

The explicit construction of $P^{\uparrow}, P^{\downarrow}$ is as follows. Each link $\ell$ in the network carries two physical attributes from the graph: a free flow speed $s(\ell)$ and a lane count $\lambda(\ell)$. Let $\bar{s}(\ell) = s(\ell)/\bar{s}$ and $\bar{\lambda}(\ell) = \lambda(\ell)/\bar{\lambda}$ denote these attributes normalized by their per-link mean over the network, and define the per-link physics weight

$$w^{\uparrow}(\ell) \;=\; \bar{s}(\ell)^{\alpha_s}\,\bar{\lambda}(\ell)^{\alpha_l}, \qquad w^{\downarrow}(\ell) \;=\; \bar{s}(\ell)^{-\alpha_s}\,\bar{\lambda}(\ell)^{-\alpha_l}, \tag{20}$$

with two real exponents $\alpha_s, \alpha_l \in \mathbb{R}$ that parametrize how strongly the chain favors fast or wide links in each phase. Each kernel is then a $w$-weighted row-stochastic walk on the directed road graph, made column-stochastic by transpose:

$$P^{\uparrow}_{j,i} \;=\; \frac{w^{\uparrow}(j)\,\mathbf{1}[\,i \to j\,]}{\sum_{j': i \to j'} w^{\uparrow}(j')}, \tag{21}$$

where $i \to j$ denotes "there is a directed road link from $i$ to $j$"; $P^{\downarrow}_{j,i}$ is defined identically with $w^{\downarrow}$ in place of $w^{\uparrow}$. The down-phase weight is the up-phase weight with $(\alpha_s, \alpha_l)$ sign-flipped, so the two phases are mirrored: when the up phase prefers slow or narrow links the down phase prefers fast or wide links, and vice versa. The default chain used throughout the evaluation has $\alpha_s = \alpha_l = 0$, which makes $w^{\uparrow}, w^{\downarrow} \equiv 1$ and both kernels reduce to the uniform-over-out-neighbors walk on the directed road graph; the speed/lane exponents come back as tuning knobs in the intervention experiment of Section 9.4.

Equivalently, ordering the $2N$ states as $\left(1^{\uparrow}, \dots, N^{\uparrow},\, 1^{\downarrow}, \dots, N^{\downarrow}\right)$, the transition matrix of the chain is the column-stochastic $2N \times 2N$ block matrix

$$M_b \;=\; \begin{pmatrix} (1-\beta)\,P^{\uparrow} & \rho\,E_b\,\mathbf{1}^{\top} \\ \beta\,I & (1-\rho)\,P^{\downarrow} \end{pmatrix}, \tag{22}$$

where $I$ is the $N \times N$ identity, $\mathbf{1} \in \mathbb{R}^N$ is the all-ones column vector, and $E_b \in \mathbb{R}^N$ is the teleportation prior treated as a column vector. The rank-1 block $\rho\,E_b\,\mathbf{1}^{\top}$ encodes the "every down-state gives mass back to the up phase via the same prior $E_b$" rule: each of its columns is the column vector $\rho\,E_b$. With this convention the chain's mass distribution $v \in \mathbb{R}^{2N}$ evolves as the simple matrix-vector product

$$v_{n+1} \;=\; M_b\,v_n, \tag{23}$$

with no transpose required at any step of the iteration.

The two scalars $\beta$ and $\rho$ control how long the chain stays in each phase before switching. A small $\beta$ lets the up-phase walk redistribute mass over many graph neighbors before committing to travel, while a small $\rho$ lets a single down-phase trip extend over many links before restarting. The values used in practice are not free parameters: they are method-of-moments estimates from the trip corpus, $\beta = \frac{1}{1+\mathbb{E}[L_{\uparrow}]}$ and $\rho = \frac{1}{1+\mathbb{E}[L_{\downarrow}]}$, where $L_{\uparrow}$ and $L_{\downarrow}$ are the empirical up- and down-phase clock lengths of the matched corpus. Their derivation from the raw matched routes is the subject of Section 8.3.

## 8.3 Calibrating $\beta$ and $\rho$ from the trip corpus

The two scalars are not tuned against the evaluation metric; they are read off the empirical distribution of trip lengths in the matched-route corpus, in three steps.

*Per-trip peak split.* Every matched route is a sequence of road links $(\ell_1, \ell_2, \ldots, \ell_K)$ produced by the Viterbi pass of Section 3.2. For each trip we identify a single *peak link* $\ell_p$ as the link with the largest score $s(\ell) = \texttt{freespeed}(\ell) \times \texttt{numLanes}(\ell)$, with ties broken by the link closest to the trip's middle. The peak link plays the role of the trip's fastest, highest-capacity segment, the freeway core of a typical commute. The two phases of the chain are then defined as the prefix and the suffix of the route relative to this peak,

$$L_\uparrow \;=\; \big|\{\, j \,:\, j < p \,\}\big|, \qquad L_\downarrow \;=\; \big|\{\, j \,:\, j > p \,\}\big|, \tag{24}$$

so that $L_\uparrow$ counts the links traversed before the peak (exclusive), $L_\downarrow$ counts the links traversed after the peak (exclusive), and the peak link itself contributes the single intermediate transition $i^\uparrow \to i^\downarrow$ of Section 8.2; in particular, $L_\uparrow + 1 + L_\downarrow \;=\; K$. By construction $L_\uparrow, L_\downarrow \in \{0, 1, 2, \ldots\}$: a route can begin or end on its own peak link, in which case the corresponding phase has length zero. Short trips ($K = 1, 2$) are handled by the convention $p = 0$ (the trip's first link is its own peak), which yields $L_\uparrow = 0$ and $L_\downarrow = K - 1$ and contributes naturally to the empirical distributions. All 714,594 matched routes are therefore included in the calibration sample.

*Empirical distribution and method-of-moments.* Figure 5 shows the histograms of $L_\uparrow$ and $L_\downarrow$ on this sample, with a geometric reference curve overlaid in black on each panel. Both empirical distributions are heavy at zero and decay roughly exponentially, which is the signature of a geometric distribution on $\{0, 1, 2, \ldots\}$ with success probability $p$ and mean $\frac{1-p}{p}$. The black curves overlay Geom($p$) with $p$ matched to the empirical mean through

$$\mathbb{E}[L] \;=\; \frac{1-p}{p} \;\Longleftrightarrow\; p \;=\; \frac{1}{1 + \mathbb{E}[L]}, \tag{25}$$

which is the method-of-moments estimator for a geometric on $\{0, 1, \ldots\}$. Read each histogram bin as "in this many trips the prefix (resp. suffix) was exactly $L$ links long." Setting $\beta = 1/(1 + \mathbb{E}[L_\uparrow])$ means the chain's average up-phase clock length matches the empirical mean, and likewise for $\rho$; the values $\beta, \rho \approx 0.10$ correspond to chains that stay in each phase about ten steps on average. Reading the empirical means off the histograms,

$$\mathbb{E}[L_\uparrow] \;=\; 8.81, \qquad \mathbb{E}[L_\downarrow] \;=\; 8.87, \tag{26}$$

and applying (25) to each phase yields

$$\beta \;=\; \frac{1}{1 + \mathbb{E}[L_\uparrow]} \;=\; 0.102, \qquad \rho \;=\; \frac{1}{1 + \mathbb{E}[L_\downarrow]} \;=\; 0.101. \tag{27}$$

These are the values fed into the chain in Sections 8.2 and 9.1; they are not chosen to fit the evaluation, only to match the average up- and down-phase clock lengths of the matched corpus. Both panels of Fig. 5 are well described by their geometric reference in the body across the full range, with the empirical bars sitting close to the geometric line throughout (a small under-dispersion at large $L$ shows up as the bars being slightly below the reference for $L_\uparrow, L_\downarrow \gtrsim 30$).

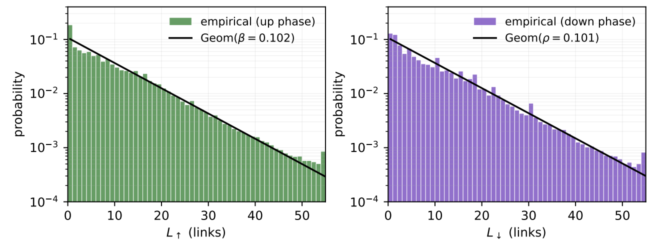


**Figure 5: Empirical histograms of $L_\uparrow$ (left) and $L_\downarrow$ (right) on the full 714,594-trip matched corpus, with the matched geometric overlay in black.**

*Why the geometric is the right reference.* In the chain of Section 8.2, the number of up-phase steps before the chain commits to travel is, by construction, a geometric random variable with success probability $\beta$ on the support $\{0, 1, 2, \ldots\}$, and the number of down-phase steps before the chain restarts is geometric with probability $\rho$ on the same support. Calibrating $\beta$ and $\rho$ by the method-of-moments (25) therefore guarantees that the chain matches the empirical mean phase length exactly. The geometric assumption is approximate (the empirical tails are slightly heavier than geometric, as visible in Fig. 5), but the two means are by construction correct, which is the moment that the chain consumes.

*Quantifying the fit.* We pair the visual fit of Fig. 5 with the Kolmogorov-Smirnov statistic $D = \sup_k \big|F_{\text{emp}}(k) - F_{\text{geom}}(k; p)\big|$, where

$$F_{\text{emp}}(k) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[\, X_i \le k \,]$$

is the empirical CDF over the $n$ observed phase lengths $\{X_i\}_{i=1}^{n}$,

$$F_{\text{geom}}(k; p) = 1 - (1-p)^{k+1}, \qquad k \in \{0, 1, 2, \ldots\},$$

is the CDF of the fitted geometric, and

$$p = \frac{1}{1 + \mathbb{E}[L]}$$

is its maximum-likelihood estimator. Smaller $D$ means a tighter empirical match. Table 7 reports $D$ in three configurations. Column $D_{\text{single}}$ fits one geometric to the full route length $K = L_\uparrow + 1 + L_\downarrow$; columns $D_\uparrow$ and $D_\downarrow$ fit geometrics to $L_\uparrow$ and $L_\downarrow$ separately. The first row matches on the low-resolution map and aggregates to low-resolution links; the second matches on the high-resolution map and aggregates to the same links; the third both matches and aggregates at high-resolution.

**Table 7: Kolmogorov-Smirnov distance $D$ between empirical phase lengths and the fitted geometric. Lower is better; read each row as "fitting a Geom($p$) to the $L_\uparrow$ (resp. $L_\downarrow$) histogram leaves a $D$ of this size under K-S."**

| Matcher | Granularity | $n$ | $D_{\text{single}}$ | $D_\uparrow$ | $D_\downarrow$ |
|---|---|---|---|---|---|
| low-res | low-res links | 293,066 | 0.188 | 0.129 | 0.103 |
| high-res | low-res links | 621,986 | 0.129 | **0.030** | **0.032** |
| high-res | high-res sub-edges | 306,659 | 0.102 | 0.138 | 0.148 |

Two findings follow. On the low-resolution aggregation that the chain runs on, splitting the route at the freespeed × permlanes peak reduces $D$ by 76% relative to the single-geometric fit, and the matching change from the low-resolution to the high-resolution map tightens $D_\uparrow$ and $D_\downarrow$ by a further 76% and 69% on the same granularity. The empirical variance moves the same way: the overdispersion ratio $\frac{\mathrm{Var}(L_\uparrow)}{\mathbb{E}[L_\uparrow](1+\mathbb{E}[L_\uparrow])}$ moves from 2.07 on the low-resolution map to 0.93 on the high-resolution map, sitting just below unity (slight under-dispersion) and well inside the geometric model's natural variance range.

At high-resolution sub-edge granularity, neither single nor split geometric is a good fit; the empirical variance is overdispersed by a factor of 1.8 to 2.0 and the up versus down split, computed at low-resolution before being projected to sub-edges, does not separate the empirical distribution cleanly. We therefore keep the geometric calibration at low-resolution ($\beta = 0.102$, $\rho = 0.101$) and treat the high-resolution view of the same data as a descriptive aggregation only.

# 9 EXPERIMENTS

The evaluation framework (Section 6) defines the metric (MRE on top-100 and overall), the three comparisons (dataset-to-dataset, forecast, reconstruct), and the 840-bin sampling protocol used by every table below.

## 9.1 Evaluation results

Table 8 reports the three comparisons in three configurations: raw prior with $\varepsilon = 10^{-6}$, diffused prior with $\varepsilon = 10^{-6}$, and diffused prior with $\varepsilon = 0$. The chain uses the calibrated parameters $\beta = 0.102, \rho = 0.101$ and $P^\uparrow, P^\downarrow$ uniform over outgoing road-graph neighbors. Iteration tolerance is $10^{-7}$, maximum 300 iterations.

**Table 8: Mean relative error of the two-phase chain across 840 random target bins (120 per day of the week) on the SLC corpus. $\beta = 0.102, \rho = 0.101$. Lower is better. "T-100" = Top-100 MRE, "Ov." = Overall MRE. "No diffusion" uses the raw prior $E_b$ of Section 4; "Diffused" uses the diffused prior $\tilde{E}_b$ of Section 5.2.**

| | No diffusion $\varepsilon=10^{-6}$ | | Diffused $\varepsilon=10^{-6}$ | | Diffused $\varepsilon=0$ | |
|---|---|---|---|---|---|---|
| Comparison | T-100 | Ov. | T-100 | Ov. | T-100 | Ov. |
| 1. dataset→dataset (next week) | 0.626 | 5.82 | 0.586 | 0.68 | 0.586 | 0.90 |
| 2. dataset→model (forecast) | 0.818 | 8.53 | 0.788 | 1.04 | 0.789 | 1.42 |
| 3. dataset→model (reconstruct) | **0.763** | **7.82** | **0.729** | **0.86** | **0.730** | **1.16** |

Three readings of the table.

*Reconstruction is comparable to forecast.* Row 3 has Top-100 MRE $\approx 0.73$, comparable to but slightly below the forecast in row 2 ($\approx 0.79$). The chain tracks the heavy-traffic head of the input only loosely: the steady state recovers the input distribution on the top-100 links to within $\sim$73%.

*Forecast sits above the noise floor.* Row 2 forecast Top-100 is $\approx 0.79$, above the noise floor of row 1 ($\approx 0.59$). The chain's forecast of next week's signal is therefore worse than last-week-at-the-same-slot. Since the chain has no temporal dynamics (no time index in $M_b$ except through the prior of the target bin), improving forecast performance would require additional signal: weekly periodicity, multi-bin context, or external covariates, none of which are part of the current model.

*Empty-link contribution to the overall MRE..* The overall MRE without the regularizer inflates to $\sim 8$ in the leftmost column because the many empty links contribute $\frac{|p_i|}{\varepsilon}$ each; $\varepsilon = 10^{-6}$ caps this contribution but does not eliminate it. With every truth value positive, the overall MRE drops to 0.67–1.37, a range in which the metric is interpretable as a relative error average. The Top-100 MRE is essentially insensitive to either the prior or $\varepsilon$, as expected: heavy-traffic links have $t_i \gg \varepsilon$ in every configuration.

## 9.2 Additive ladder: damping and road-type tuning on the whole network

This subsection isolates the contributions of the damping rate and the road-type weighting in two stages on the same evaluation sample. The sample is a set of 840 calendar pairs $(t,\ t{+}7\,\mathrm{d})$, drawn so that exactly 120 pairs sit on each day of the week (seed 7), with $t$ chosen from the first 23 days of the corpus so that the next-week sibling is in range. The model receives $\tilde{E}_b(t)$ and predicts $\tilde{E}_b(t{+}7\,\mathrm{d})$. The per-pair error is the Overall MRE (over all $N$ links), and we report its mean across the 840 pairs. Parameter bounds follow the geometric-fit discussion above: $\alpha,\ \beta,\ \rho \in [0.01,\ 0.15]$, while $\alpha_s$ and $\alpha_l$ are unconstrained.

*Damping sweep (no road-type tuning).* We first hold $\alpha_s = \alpha_l = 0$ and sweep the damping rate. For the single-phase chain the rate is $\alpha$. For the two-phase chain we set $\beta = \rho$ and sweep the common value, so that the two columns differ only by the chain structure.

**Table 9: Damping sweep on the whole network, 840 pairs $(t,\ t{+}7\,\mathrm{d})$, no road-type weighting. Overall MRE (lower is better); bold marks the best in each column.**

| rate | Single-phase ($\alpha$) | Two-phase ($\beta{=}\rho$) |
|---|---|---|
| 0.03 | 1.130 | 1.181 |
| 0.04 | 1.098 | 1.154 |
| 0.05 | 1.072 | 1.131 |
| 0.06 | 1.049 | 1.110 |
| 0.07 | 1.029 | 1.091 |
| 0.15 | **0.923** | **0.984** |

Both columns improve monotonically as the damping rate increases toward the upper bound of the search box. Single-phase outperforms two-phase at every common rate by approximately 0.05–0.06 on Overall MRE.

*Road-type tuning on top.* Table 10 fixes the damping rate at the upper bound and adds road-type weighting, either lanes only, speed only, or both jointly. Each tuned operating point is found by L-BFGS-B with finite-difference gradients and five random starts; the optimization minimizes Overall MRE on the same 840 pairs. The vanilla rows from Table 9 are reproduced for reference at the same damping rate.

**Table 10: Road-type tuning on the 840 pairs, Overall MRE objective. Single-phase block: $\alpha \in [0.01, 0.15]$, $\alpha_s, \alpha_l$ unbounded unless noted. Two-phase block: $\beta, \rho \in [0.01, 0.15]$, $\alpha_s, \alpha_l$ unbounded unless noted. Vanilla rows are objective-independent; tuned rows report the operating point at the optimizer's solution.**

| variant | parameters | Ov. MRE |
|---|---|---|
| *Single-phase* | | |
| vanilla | $\alpha=0.05$ | 1.072 |
| vanilla | $\alpha=0.15$ | 0.923 |
| + lanes only | $\alpha=0.150$, $\alpha_l=+0.337$ | 0.916 |
| + speed only | $\alpha=0.150$, $\alpha_s=+2.859$ | 0.870 |
| + both | $\alpha=0.150$, $\alpha_s=+2.674$, $\alpha_l=+0.169$ | 0.869 |
| + both, $\alpha$ fixed | $\alpha=0.05$; $\alpha_s=+2.774$, $\alpha_l=+0.243$ | 1.002 |
| *Two-phase* | | |
| vanilla | $\beta=0.102$, $\rho=0.101$ | 1.041 |
| vanilla | $\beta=\rho=0.15$ | 0.984 |
| + lanes only | $\beta=0.150$, $\rho=0.150$, $\alpha_l=+1.875$ | 0.962 |
| + speed only | $\beta, \rho, \alpha_s$ tuned | — |
| + both | $\beta, \rho, \alpha_s, \alpha_l$ tuned | — |
| + both, $(\beta, \rho)$ fixed | $(\beta, \rho) = (0.102, 0.101)$; $\alpha_s, \alpha_l$ tuned | — |

*Discussion.*

## 9.3 Baselines and alternative pipelines

To show that our model is doing real work, we compare it against simpler pipelines. Each simpler pipeline drops one part of our method: the up and down split, the diffusion smoothing, the high-resolution map, or the model itself. We feed each pipeline the same data and measure the same three errors as Table 8. If our full model beats every simpler version, each part of the method earns its place.

We test three simpler pipelines:

- **Naive.** Copy last week's counts as the prediction for this week. No model, no smoothing. This is already the dataset-to-dataset row of Table 8 and serves as the noise floor.
- **One-phase PageRank.** A single PageRank chain on the same prior, with no split into an up phase and a down phase. This tests whether the two-phase decomposition of Section 8.1 actually helps.
- **Low-resolution matching.** Keep the full two-phase chain but build the popularity counts from matching on the low-resolution map instead of the high-resolution one. This is the version of the pipeline used before the matching upgrade in Section 3.2, and tests whether the long curvy roads recovered by the high-resolution matcher matter for prediction quality.

Each baseline runs on the same 840 bins as the main evaluation and reports the same three errors.

## 9.4 Intervention case study: Utah vs Washington, Sept 15 2018

On Saturday September 15, 2018 at 8:00 PM MDT the University of Utah hosted the University of Washington at Rice-Eccles Stadium (40.7596° N, 111.8485° W), with reported attendance of 47,445. The game lies in the middle of our 30-day corpus, so we can ask whether the chain's pre-kickoff-hour forecast on this particular Saturday differs in a measurable way from a normal Saturday at the same time of day, and whether tuning the chain's parameters on this event window recovers any of that gap.

*Cluster construction.* We rank road links by their typical pre-kickoff traffic and keep the busiest few near the stadium. Concretely: take every link whose midpoint lies within 1 km of Rice-Eccles, which leaves 305 candidate links. Each candidate carries a per-bin link probability $E_b(i)$ summing to one over the full network; we average those per-bin probabilities across the kickoff hour (12 five-minute bins, 19:00 to 19:55) on three clean baseline Saturdays (Sept 1, 8, and 22), giving $12 \times 3 = 36$ probability values per link that we collapse to one scalar "mean kickoff-hour popularity" by averaging. (Sept 15 is the event itself, and Sept 29 is another Utah home football game, so neither is used.) We then take the top-$k$ candidates by that scalar. We report $k \in \{10, 20, 30\}$. The $k = 30$ cluster is shown in Fig. 6: 30 red links concentrated along the east-west arterial spine immediately south of the campus, plus a couple of north-south cross-streets.

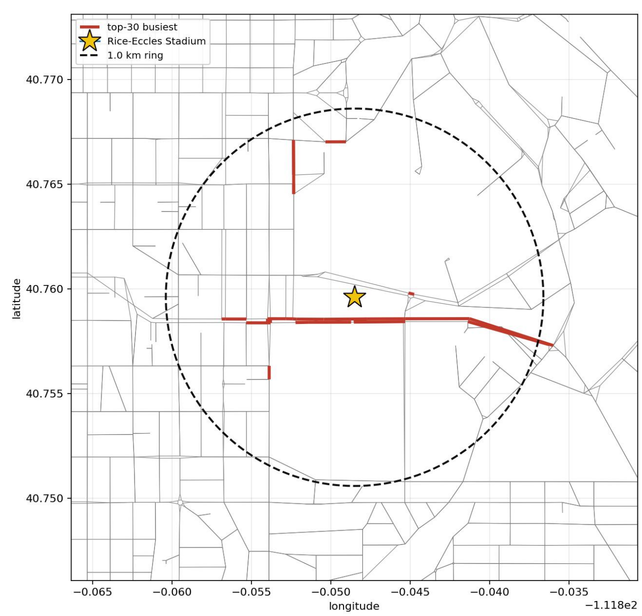


**Figure 6: Cluster ranked over 19:00–19:55 on Sept 1, 8, and 22.**

*Experimental design.* The chain's parameters are trained on non-event Saturday pairs at the same kickoff hour (Sept 1 → Sept 8 and Sept 22 → Sept 29, 19:00–19:55) and then frozen. At test time the frozen chain is given the *trip-origins distribution* of Sept 15, the per-link per-bin count of trips that start on each link during the kickoff hour, normalized to a probability. The origins vector carries the event signal (extra starts around the stadium) without leaking the link-popularity values we are predicting. Parameters never see Sept 15 link densities. The output of the frozen chain on the Sept 15 origins input is compared against Sept 15 link popularity on the top-$k$ cluster; lower is better. Both single-phase and two-phase chains are tuned with the same L-BFGS-B protocol on the training pairs (five multi-start initializations, $p, \beta, \rho \in [0.01, 0.15]$, $\alpha_s, \alpha_l$ unbounded).

*Headline result: held-out comparison.* Table 11 reports the held-out d2m on the Sept 15 cluster for the proposed two-phase chain, the single-phase one-stage Markov chain baseline, and three simpler predictors that drop one component of the pipeline.

**Table 11: Held-out forecast d2m on the Sept 15 stadium cluster. Parameters fit only on the non-event Saturday pairs (Sept 1 → Sept 8 and Sept 22 → Sept 29 at the kickoff hour) and frozen before evaluation on Sept 15. Test input is the Sept 15 trip-origins distribution after $\gamma$-diffusion. Lower is better. Read each cell as "this predictor's mean per-link relative error, averaged over the 12 kickoff-hour bins and the top-$k$ stadium links."**

| predictor | top-10 | top-20 | top-30 | notes |
|---|---|---|---|---|
| two-phase, tuned (ours) | **0.678** | **0.751** | **0.848** | main method |
| single-phase, tuned | 0.709 | 0.826 | 0.852 | 1-stage Markov chain |
| two-phase, no dangling fix (v1) | 1.734 | 2.160 | 2.054 | earlier two-phase variant |
| two-phase, no $\gamma$-smoothing | 1.785 | 2.249 | 2.084 | raw Laplace prior |
| *d2d* (last week's popularity, no chain) | 7.088 | 8.918 | 8.062 | naive baseline |
| climatology (non-event Saturday mean) | 17.51 | 14.01 | 11.39 | worst predictor |

*Reading the table.* The two-phase chain is the best predictor on every cluster under the held-out protocol, by 0.031 d2m on top-10, 0.075 d2m on top-20, and 0.004 d2m on top-30 relative to the single-phase chain. The remaining baselines tell a consistent story: the naive *d2d* row uses last week's popularity and lands at the event-day noise floor; climatology averages over non-event Saturdays and is the worst predictor by far because the stadium cluster is by definition event-driven; the v1 chain without the dangling-link fix and the no-$\gamma$-smoothing variant both sit well above the tuned chains, showing that the dangling correction and the $\gamma$-smoothing component each earn their place in the pipeline.

**Table 12: Frozen-tuned parameter values per cluster, fit on the non-event Saturday training pairs (origins → popularity). Parameter ranges: $p, \beta, \rho \in [0.01, 0.15]$, $\alpha_s, \alpha_l$ unbounded. Read each row as "on this cluster size, L-BFGS-B converged to these parameters and produced the d2m bolded in Table 11; $\alpha_s, \alpha_l$ are unbounded exponents and can fall outside $[-1, 1]$."**

| | single-phase | | | two-phase | | | |
|---|---|---|---|---|---|---|---|
| cluster | $p$ | $\alpha_s$ | $\alpha_l$ | $\alpha_s$ | $\alpha_l$ | $\beta$ | $\rho$ |
| top-10 | 0.0100 | −2.39 | +4.67 | +3.62 | −4.22 | 0.1500 | 0.0100 |
| top-20 | 0.0100 | −4.11 | −0.62 | −6.43 | −1.03 | 0.0100 | 0.1049 |
| top-30 | 0.0100 | +47.36 | −15.03 | +77.20 | −33.47 | 0.0100 | 0.0100 |

## 10 DISCUSSION AND LIMITATIONS

*What the held-out experiment shows.* Under proper held-out conditions the two-phase chain remains the best predictor on the Sept 15 stadium cluster across the predictors we considered: it beats the single-phase chain, the v1 two-phase chain without the dangling-node fix, the no-$\gamma$-smoothing variant, and the no-chain baselines. The gap between the two-phase chain and the single-phase chain is smaller than under the in-sample protocol because the held-out setup denies both chains direct access to the event-day ground truth, but the ordering is preserved across all three cluster sizes (top-10, top-20, top-30). Every component of the pipeline (Laplace pseudo-count, $\gamma = 0.20$ diffusion, dangling-node correction, two-phase split) is shown to earn its place by an ablation against the corresponding simpler variant in Table 11.

*Limitations.* The probe corpus is a biased sample (Section 4): it oversamples freeway corridors and undersamples residential streets. The chain therefore predicts *probe-population link popularity*, not a uniform-vehicle volume. The validation also tests prediction at a single time scale (5-minute bins) and a single city; cross-city transfer and finer-grained resolutions are outside scope. Finally, the intervention case study uses a single known event; an evaluation across many labeled events would strengthen the generalization claim.

## 11 CONCLUSION

We presented a two-phase PageRank chain that turns a smoothed per-link popularity prior into a per-link traffic prediction on a 99,716-link city graph, with all calibration steps ($\sigma$ for map-matching, $\gamma$ for diffusion smoothing, $\beta$ and $\rho$ for the chain) driven by the data. Under a proper held-out protocol on a real event (Utah vs Washington, Sept 15 2018), the two-phase chain beats the single-phase baseline and a suite of simpler comparators on every stadium cluster size, validating both the two-phase architecture and the calibration pipeline.

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